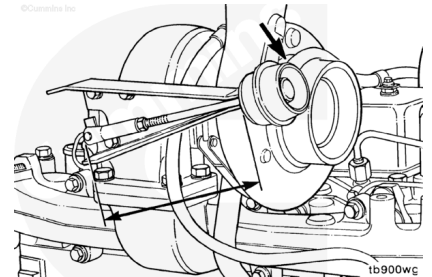


Trainings ▾

Remove

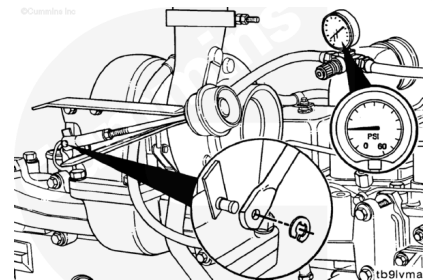
Note : Prior to removal, note the position of the control rod from the boost capsule housing and orientation of the boost capsule hose connector in relation to the mounting bracket.



Remove the retaining clip from the control lever.

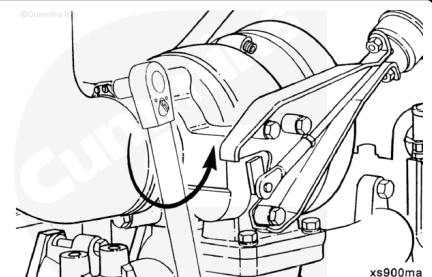
Disconnect the boost capsule actuator rod end from the turbocharger wastegate lever. This can be accomplished by applying regulated air pressure to the boost capsule in a sufficient amount to activate control rod movement.

Disconnect the control rod from the turbocharger wastegate lever pin.



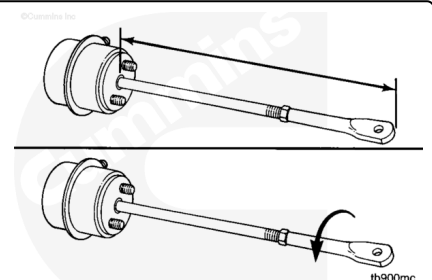
Note : If the boost capsule diaphragm material is ruptured and will not hold air pressure other than by hand, manually pull the control rod outward to overcome boost capsule spring tension for removal of the control rod from the turbocharger wastegate lever pin.

Loosen the boost capsule mounting capscrews, disconnect the air supply hose, and remove assembly from the mounting bracket.



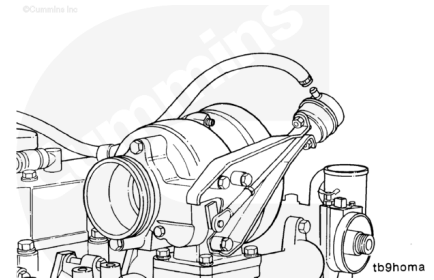
Record the length of adjusting link prior to removal.

Loosen the nut and remove the adjusting link end from the boost capsule actuator.



Inspect for Reuse

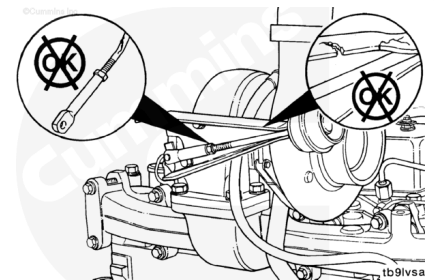
Inspect the wastegate actuator hose for cracks or holes.
Replace the hose if damaged.



Inspect the wastegate mounting bracket, actuator rod, and lever for damage.

A bent wastegate mounting bracket, actuator rod, or lever can cause improper operation.

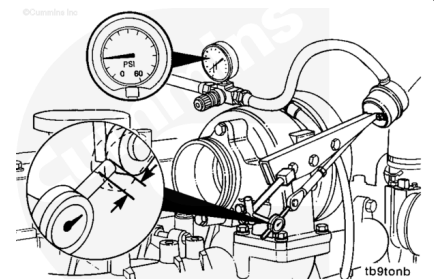
If the wastegate mounting bracket, actuator rod, or lever is bent, it **must** be replaced.



Pressure Test

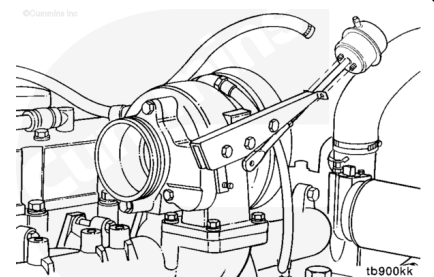
Connect regulated air pressure and an accurate pressure gauge to the capsule. Apply 180 kPa [26 psi] to make sure the wastegate is functioning properly. The actuating rod **must** move approximately 5 mm [0.2 in] without any sticking or air leakage.

Note : A small amount of travel when air pressure is first applied is normal; the tolerance is being removed from the system.



Replace the actuator if no movement of the actuator rod and lever is detected.

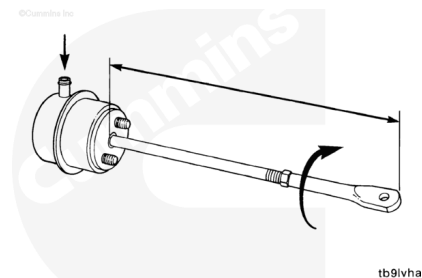
If repair is **not** correct, refer to Turbocharger Wastegate Actuator - Calibration in this section.



Install

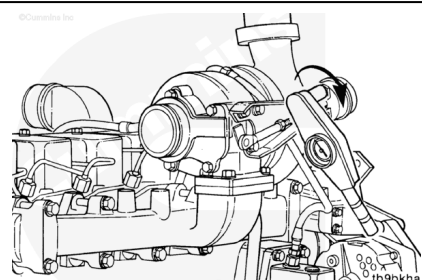
Install the adjusting link end onto the boost capsule actuator assembly.

Adjust the rod to approximately the same length as when removed.



Fit the new boost capsule actuator assembly to the actuator mounting bracket, and install the mounting capscrews.

Torque Value: 4.5 n•m [40 in-lb]

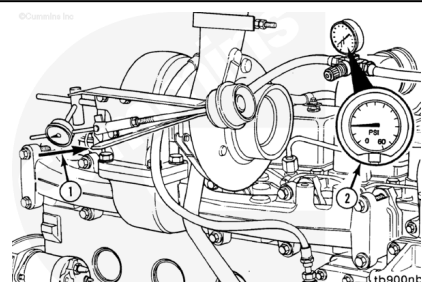


Calibrate

Note : The wastegate is set accurately from the factory. Adjustment is not necessary unless the capsule is removed.

Connect clean, regulated air pressure to the boost capsule.

Regulate the air pressure to the wastegate capsule according to the following table.



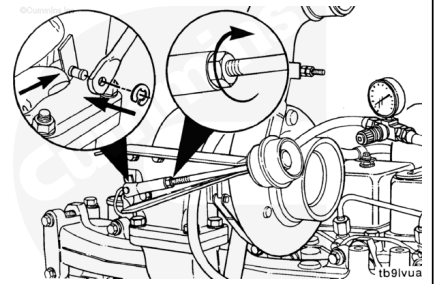
Model Year	Rating (hp)	Application	Pressure (kPa)	Pressure (psi)
'96	190 to 230	Automotive	198	29

Measure the actuator travel:

mm		in
0.33	MIN	0.013
1.30	MAX	0.051

⚠ CAUTION ⚠

Do not pull, push, or force alignment of the clevis pin. This can damage the wastegate.



Adjust the wastegate, if necessary, to achieve specified travel.

- Pull the wastegate lever to the foremost closed position (lever toward boost capsule).
- Adjust the length of the clevis end of the control rod to where the clevis pin hole aligns to the wastegate lever.
- Install the adjusting link and retaining clip.
- After adjustment is complete, tighten actuator rod jam nuts.

Disconnect the regulated air pressure line from the boost capsule.

Connect the turboboost line to the boost capsule, and secure with hose clamps.

Note : If possible, a more accurate method of wastegate adjustment is to inspect the manifold pressure at rated rpm.

